Training Day 18

1. Topics Covered and Brief Explanation

1. Introduction to Computer Vision: Computer Vision is a field of Artificial Intelligence that enables computers to understand and interpret images and videos.
2. How Computers Understand Images: Images are represented as collections of pixels. Each pixel contains numerical values that represent colour and intensity.
3. Types of Images:
    - Grayscale Images: Contain different shades of grey.
    - Colour Images: Usually contain three colour channels: Red, Green, and Blue (RGB).
4. Image Processing: It involves modifying or analysing images to improve their quality or extract useful information.
5. Basic Computer Vision Tasks:
    - Image classification – identifying the category of an image.
    - Object detection – identifying and locating objects.
    - Face detection – detecting human faces in images.
    - Image segmentation – dividing an image into meaningful regions.
6. Applications of Computer Vision: Used in facial recognition, medical imaging, traffic monitoring, security systems, autonomous vehicles, and object detection.

2. Tools and Technologies Used

Tool/Technology	Purpose
Python	Programming language
OpenCV	Image processing and computer vision
NumPy	Handling image data as arrays
Google Colab / VS Code	Writing and executing code

3. Practical Learning / Code Examples

A basic example was used to understand how an image can be loaded and displayed using OpenCV.

```
import cv2

# Load an image

image = cv2.imread("image.jpg")
```

```
# Check whether the image was loaded
if image is not None:
    print("Image loaded successfully")
    print("Image dimensions:", image.shape)
else:
    print("Image not found")
```

Explanation:

- cv2.imread() loads an image from the specified path.
- image.shape displays the image dimensions in terms of height, width, and colour channels.
- The condition checks whether the image was loaded successfully.

4. Learning Outcomes

- Understood the fundamentals of Computer Vision.
- Learned how computers represent images using pixels.
- Differentiated between grayscale and colour images.
- Explored basic computer vision tasks.
- Identified real-world applications of Computer Vision.